



RENEWABLE ENERGY: SOLAR OVEN FUN!

Let's Use Cooking Magic from Nature!

Grade 3 Science & Environment Lesson Guide

 **Cook delicious snacks using only the heat of the sun!**

No gas, no matches, and no electricity needed!



Feeling the Heat from the Sun!

Have you ever stepped inside a car left parked outside on a hot summer afternoon? 🚗

Whew! It feels like a boiling room inside, doesn't it? Or think about walking barefoot on your terrace floor at noon—ouch! It burns your feet because the surface gets so intensely hot.

Why does it get so incredibly hot inside a closed car? It happens because the sun shines straight through the windows, and the heat rays get completely locked inside with nowhere to escape!

This is exactly how a science project called a ****Solar Oven**** works! It behaves like a small, friendly trap that catches warm sunbeams and uses them to heat up things for us.

Turn and Share:

Turn to your classroom neighbor right now and share a memory: where have YOU felt the sun's warm heat the absolute strongest in your life?

What Is Renewable Energy?

Before we look at our special oven, let's learn a big science word: **Renewable Energy** (say: *Re-new-a-ble En-er-gy*).

Renewable energy simply means clean power that comes straight from nature and **never runs out**, no matter how much we use it! The sun is our biggest helper for this.

Think about a smartphone battery or a gaming toy. After you play with it for a few hours, the battery runs completely flat. It goes to sleep and needs to be plugged into a wall socket to recharge.

But what about the mighty sun? It shines bright every single morning without ever needing a plug, a wire, or a recharge! The sun is completely **free** and stays with us **forever**.



Saving the Planet:

Because the sun is a natural powerhouse, using its light and heat to do chores—like cooking our lunch—is amazing for the Earth. It doesn't create smoke, so it keeps our environment beautiful and clean!



Meet the Solar Oven's 3 Magic Parts!

A solar oven is just a cardboard box fitted with three special science helpers. Each helper has an important job to do:



1. The Aluminium Foil Flap — "The Mirror Friend"

This shiny top flap acts like a mirror. It catches passing sunbeams and bounces them straight down into the box to build up internal warmth!



2. The Plastic Wrap Cover — "The Magic Trapdoor"

This clear see-through sheet lets bright sunlight go down inside, but acts like a heavy blanket that keeps all the lovely heat trapped safely inside!



3. The Black Paper Lining — "The Heat Sponge"

This dark paper layer sits on the bottom of the box. It works like an active sponge, soaking up every bit of light energy to heat your food quickly!

Bouncing Sunlight with a Steel Thali!

Let's find out how our **Mirror Friend** works. Have you ever taken a shiny steel thali plate out into the sunny yard or balcony?

If you tilt the thali just right, it reflects a bright patch of light onto the wall! When light hits a super shiny, smooth surface, it can't go through it. Instead, it bounces away immediately—just like a tennis ball bouncing off a concrete wall!

In science, this bouncing trick is called **Reflection** (say: *Re-flek-shun*).

 **"The steel thali bounces light just like the Aluminium Foil flap in our solar oven!"**

By angling our shiny foil flap toward the sun, we gather extra sunbeams that would have missed our box and direct them straight inside to multiply the heat.

The Heat Sponge: Dark Colors Soak Up Heat!

Have you ever noticed that if you wear a dark black sweater or t-shirt outside in the hot sun, you start feeling sweaty very quickly?

This happens because different colors treat light differently. Light, bright colors like white act like shields and reflect light away. But ****dark colors, especially black****, act like magnets for sunbeams! They absorb the light and turn it into pure heat.

That is why we line the interior bottom of our solar oven with dark black paper. It acts as our reliable ****Heat Sponge****!



Touch Experiment to Try:

Next time you are outside on a sunny afternoon, gently touch a white stone or car, and then touch a black one. Which one feels much hotter against your fingers? The dark one always wins!

The Magic Window Trap!

Now, how do we make sure our captured heat doesn't just float away back into the air? That is where the clear plastic wrap—our ****Magic Trapdoor****—comes to save the day!

Sunbeams travel in a special way that lets them pass straight through clear plastic wrap without any trouble. They fly inside, hit our black heat sponge, and transform into warm air waves.

But here is the trick: warm air waves are too large and sluggish to pass back out through the plastic wrap! They get blocked completely.



"The plastic wrap works just like a warm winter blanket for the sun's captured rays!"

Because the clear wrap keeps letting light inside while sealing all the hot air tightly within the box, the temperature climbs higher and higher until it is ready to bake!

Mini-Chef Recipe Time!

Let's do some solar cooking! Here are two fun treats you can make:

Recipe 1: Cheesy Sun-Toast

Place a slice of cheese neatly onto a piece of bread. Put it inside your oven tray and watch the cheese turn soft, gooey, and beautifully melted!

Recipe 2: Chocolate Biscuit S'mores

Take two round Marie biscuits and drop a few chocolate chips on top of them. Set them in the solar box and wait for the sweet chocolate chips to melt into delicious cream!

Important Rule for Eco-Chefs:

Only use natural sunlight today! No kitchen gas stoves, no electrical appliances, and absolutely no matches are allowed. The sun will do all the heavy work for us!



Eco-Hero Choices: What's Safe?

To make sure your science experiments are always happy and successful, look at these three simple guidelines:

- **DO: Use your oven on bright, clear days!**

The absolute best time is a sunny Sunday noon when the sun is directly overhead and shining its strongest beams. More sunshine equals more heat!

- **DON'T: Put plastic toys or bags inside!**

Toy items are not meant for cooking. They will melt down into sticky plastic goo and ruin your project. Only use food items like biscuits, bread, or cheese.

- **DON'T: Cook on cloudy or monsoon days!**

When the sky is dark and rainy, the sun hides behind gray clouds. There won't be enough solar energy coming through to heat up your snacks. Save your cooking for a clear day!

Summary & The Eco-Chef Pledge!

Before we head out to catch some sunshine, let's repeat our three quick science rules for the solar oven:

 Foil Flap	Our Mirror Friend that bounces and reflects light down.
 Plastic Wrap	Our Magic Trapdoor that seals the warm air inside.
 Black Paper	Our Heat Sponge that absorbs sunbeams instantly.

Super Crucial Safety Rules:

- **Never look directly at the bright sun!** It is way too powerful and can hurt your eyes.
- **Always experiment with an adult!** Solar cooking is fun and perfectly safe when you have grown-up help around.



Thank you for learning with us! Keep exploring science!